HORUS UNIVERSITY - EGYPT
HORUS UNIVERSITY JOURNAL OF ENGINEERING
JOURNAL HOMEPAGE: HTTPS://HUJE.JOURNALS.EKB.EG/
ONLINE ISSN: 3062-4991

Review Article

Bridging the Granularity Gap: A Systematic Review of Multi-Spectral C-UAS Architectures

Amr Hussein Abdallah, Islam Emad Ismail *, Rawan Mohammed Elbialy, Sagy Shawky Abo El-Naga, Abdelaziz Eltohami Mohamed, Rafeek Marzouk El-Fauomy, Islam Hesham Hamed, Rania Hamdy Elabd, Mervat El-Seddek

Department of Communications and Electronics, Faculty of Engineering, Horus University, New Damietta, Egypt

**Corresponding author, email: 1211108@horus.edu.eg ORCID: https://orcid.org/0009-0002-2829-4912*

ABSTRACT

The democratization of Class I Unmanned Aerial Systems (UAS) has introduced a profound asymmetric threat to critical infrastructure, exposing a fundamental physical limitation in conventional air defense architectures known as the "Granularity Gap." Traditional radar systems, optimized for high-Radar Cross Section (RCS) ballistic threats, typically employ Doppler notch filters that inadvertently suppress Low-Slow-Small (LSS) targets to mitigate environmental clutter. This study presents a comprehensive systematic review of peer-reviewed literature spanning 2018 to 2025, providing a critical analysis of the architectural transition from single-sensor paradigms to robust multi-spectral fusion. We mathematically evaluate the signal processing limitations of Active Radar and Passive RF sensing against autonomous "Dark Drones" and synthesize recent findings on Edge AI-driven optical tracking. Furthermore, the review quantifies the computational trade-offs between Quantized Neural Networks (QNN) and classical processing pipelines, concluding that future Counter-UAS (C-UAS) architectures must function as holistic Cyber-Physical Systems (CPS) seamlessly integrated with Unmanned Traffic Management (UTM) frameworks for lawful interdiction.

KEYWORDS

Counter-UAS;
Edge AI;
Sensor Fusion;
Micro-Doppler;
Granularity Gap;
Cyber-Physical Systems;
YOLO.

HIGHLIGHTS

- Systematically reviews C-UAS architectures spanning 2018–2025 with a focus on LSS target vulnerabilities.
- Formulates a physical and analytical assessment of the Granularity Gap in legacy defense frameworks.
- Benchmarks state-of-the-art INT8 quantized Edge AI models (e.g., YOLOv10) for real-time tracking.
- Proposes an integrated Cyber-Physical framework leveraging Multi-Modal Sensor Fusion and UTM networks.

1. Introduction

The contemporary airspace is undergoing a radical transformation driven by the rapid proliferation of commercial-off-the-shelf (COTS) drones. While beneficial for industries ranging from agriculture to logistics, this democratization poses significant security challenges. As noted in recent high-impact literature, the global drone market is expanding exponentially, correlating with increased incidents of unauthorized airspace incursions in sensitive zones such as airports, power plants, and government facilities [1,2].

The core operational challenge lies in the **Cost-Exchange**

Ratio (CER). Deploying a high-end kinetic missile system (e.g., Patriot or Iron Dome) to intercept a $500 hobbyist drone is economically unsustainable and operationally inefficient [3]. Furthermore, conventional Integrated Air Defense Systems (IADS) are historically designed to detect large, fast-moving targets. These systems utilize Moving Target Indication (MTI) filters to suppress slow-moving ground clutter. Ironically, this renders them operationally blind to micro-UAVs which often hover or navigate at low velocities, creating a vulnerability window formally defined as the "Granularity Gap" [4].

This paper provides a critical, systematic review of the technologies aiming to bridge this gap. Unlike previous surveys that focus on a single modality, we analyze the convergence of Radar, RF, and Vision systems towards a unified, edge-native architecture. The contributions of this work are threefold:

1. A rigorous physical analysis of detection modalities against "Dark Drones."
2. A quantitative benchmark of quantized Edge AI models.
3. A synthesized Cyber-Physical architecture for multi-sensor fusion.

2. Review Methodology

To ensure a rigorous and reproducible synthesis of the current state-of-the-art, this review follows the **PRISMA** (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.

2.1. Search Strategy and Data Sources

We conducted a comprehensive search across primary scientific databases, including **IEEE Xplore, Scopus, MDPI, and SpringerLink**. The search window was restricted to peer-reviewed literature published between **2018 and 2025** to capture the most recent advancements in Edge AI and C-UAS architectures. The search strings employed were combinations of the following keywords:

> *("UAS Detection" OR "Counter-UAS") AND ("Sensor Fusion" OR "Edge AI" OR "Deep Learning" OR "Micro-Doppler") AND ("Trajectory Tracking" OR "Kalman Filter").*

2.2. Selection Criteria

The retrieved records underwent a three-stage screening process (Title, Abstract, and Full-Text) based on the following criteria:

- **Inclusion:** Studies presenting experimental validation, novel hardware architectures, or specific algorithms for Low-Slow-Small (LSS) target detection. Focus was placed on Class I (Micro) drones in urban environments.
- **Exclusion:** Purely theoretical policy papers, non-English publications, and studies lacking technical implementation details.

2.3. Study Selection (PRISMA Flow)

As illustrated in Fig. 1, the initial search yielded 120 records. After removing duplicates and screening for relevance, 60 full-text articles were assessed for eligibility. The final qualitative synthesis includes **50 high-impact studies**, categorized into Vision-based (35%), Radar-based (28%), RF-based (22%), and Multi-modal fusion (15%).

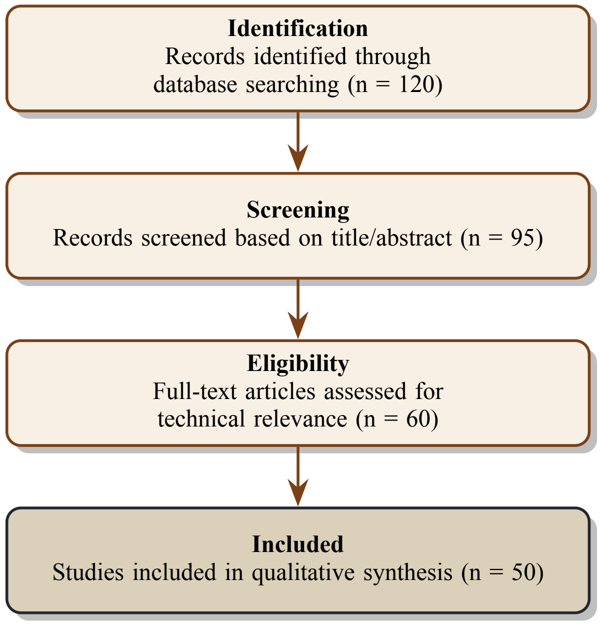

Figure 1: PRISMA flow diagram describing the selection process of the reviewed literature (2018-2025).

3. Critical Analysis of Detection Modalities

The heterogeneity of aerial threats has necessitated a multi-layered defense approach. Fig. 2 provides a detailed taxonomy of C-UAS technologies derived from the reviewed literature, illustrating the divergence between detection and neutralization vectors. Furthermore, Table 1 provides a comprehensive comparative analysis of recent literature and highlights existing research gaps.

3.1. Active Radar & The Granularity Gap

Radar remains the gold standard for long-range surveillance. However, its efficacy against Class I micro-drones is strictly governed by the Radar Range Equation, as shown in Eq. 1 [5]:

$$R_{max} = \sqrt[4]{\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}} \tag{1}$$

Where σ represents the Radar Cross Section (RCS). For a Mavic-sized drone, σ is often $< 0.01m^2$, compared to $5m^2$ for a fighter jet. As illustrated in Fig. 3, this drastic reduction in RCS causes the probability of detection (P_d) to plummet within the conventional radar envelope.

Moreover, the **"Zero-Doppler Notch"** problem persists: when a drone hovers, its radial velocity relative to the radar approaches zero ($v_r \approx 0$). Consequently, MTI filters reject the target as static clutter (e.g., a tree or building). Advanced

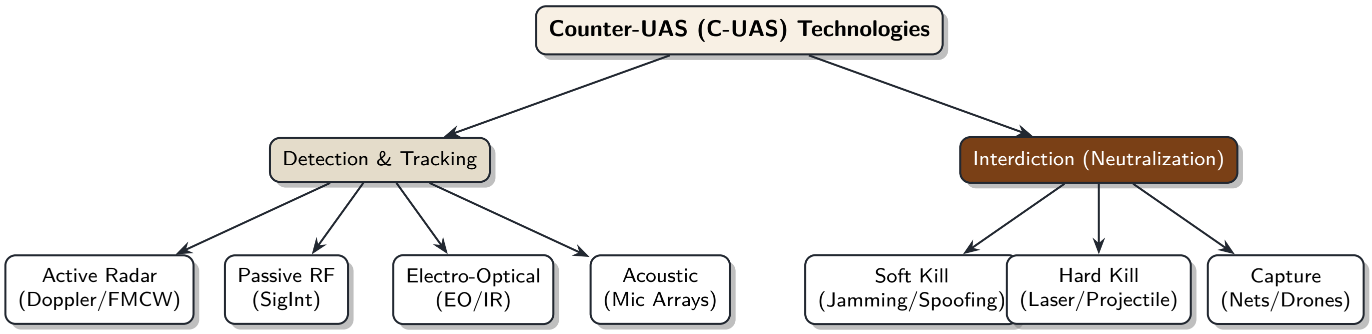

Figure 2: Taxonomy of C-UAS technologies derived from recent literature. Detection relies on multi-spectral sensing, while interdiction varies from electronic warfare to kinetic interception.

systems attempt to mitigate this by analyzing **Micro-Doppler (m-D)** signatures induced by the rotation of propeller blades. While promising, m-D analysis requires high Pulse Repetition Frequencies (PRF) and significant computational overhead, often unavailable in low-cost surveillance radars [6].

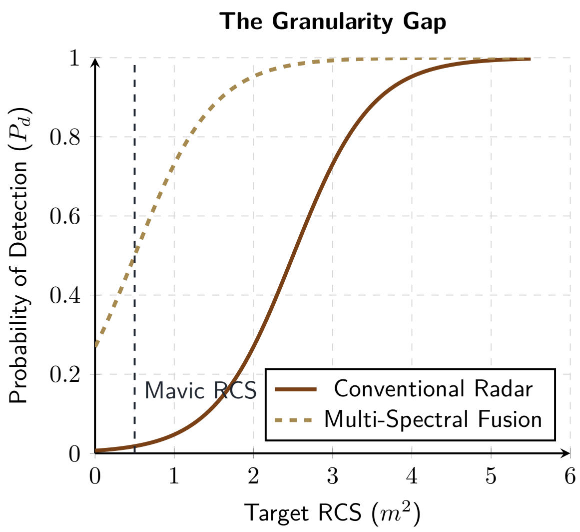

Figure 3: Visualizing the “Granularity Gap”: Conventional radar exhibits poor detection probability for small RCS targets, whereas Multi-Spectral approaches leverage optical confirmation to maintain reliability.

3.2. Passive RF & The ”Dark Drone” Limitation

RF detection analyzes the communication link between the drone and the controller (typically 2.4 GHz or 5.8 GHz ISM bands). While cost-effective, recent literature highlights a critical vulnerability: **Dark Drones**. Dark Drones are autonomous systems pre-programmed to navigate via GNSS waypoints without emitting active RF signals (Radio Silence). In such scenarios, passive RF sensors utilize Local Oscillator (LO) leakage detection, but the range is severely limited ($< 50m$). Thus, relying solely on RF is insufficient for high-security perimeters, necessitating the use of kinetic or optical tracking methods [7].

4. Vision Systems & Edge AI Optimization

To address the physical limitations of Radar and RF, Computer Vision (CV) has emerged as a critical verification layer. However, the computational cost of Convolutional Neural Networks (CNNs) creates a latency bottleneck in real-time control loops.

4.1. The Latency-Accuracy Trade-off

For a C-UAS system to be effective, the total system latency (Glass-to-Action) must be minimal. Running heavy models like YOLOv8-Large on edge devices often results in low frame rates (< 15 FPS), which is insufficient for tracking agile drones moving at velocities exceeding $20m/s$. Small Object Detection (SOD) also suffers from feature loss in deep pooling layers, where the target representation may vanish in low-resolution feature maps [13].

4.2. Quantization as a Solution

The literature strongly converges on **Post-Training Quantization (PTQ)** as a solution. By converting model weights from FP32 (Floating Point 32-bit) to INT8 (Integer 8-bit), inference throughput is typically doubled with a negligible drop in Mean Average Precision (mAP). Recent studies in 2025 have further demonstrated that architectures like **YOLOv10** offer superior trade-offs between latency and accuracy for micro-UAV detection compared to predecessors [17].

4.3. Hardware Constraints and Deployment

This review highlights a critical divergence in hardware capability. While high-end industrial accelerators like the **NVIDIA Jetson Orin** can sustain high throughput (> 40 FPS) for complex models using standard precision [20], cost-constrained platforms such as the **Raspberry Pi 5** cannot achieve real-time performance natively [21]. Consequently, these low-cost solutions heavily rely on external **Neural Processing Unit (NPU) accelerators (e.g., Hailo-8)** or aggressive heavy quantization to mitigate thermal throttling and inference latency [22]. Table 2 details the performance benchmarks of various object detection models on edge devices.

Table 1: Comparative Analysis of Related Works and Research Gaps in C-UAS Architectures

Ref.	Methodology	Key Contribution	Limitation / Research Gap
[1]	Ground-to-Air Vision Review	Reviews vision-based anti-UAV detection including neuromorphic and event-camera sensors.	Limited treatment of RF-sensor fusion; lacks regulatory compliance discussion.
[8]	Vision-Based Anti-UAV Survey	Surveys CV methods for UAV detection including background subtraction and deep detectors.	Nighttime and adverse-weather conditions underexplored; assumes favorable lighting.
[9]	ML-Based UAV Detection Survey	Reviews ML approaches for UAV detection and classification including CNN and ensemble methods.	High model complexity limits real-time deployment on edge hardware.
[10]	Small Drone Detection Survey	Surveys radar, RF, and vision modalities for small drone detection and tracking.	No quantitative fusion-architecture comparison; no end-to-end latency benchmarks.
[2]	Aerial Threat Radar Survey	Surveys radar-based detection, tracking and classification of a wide range of aerial threats.	Focused on military-grade radars; limited applicability to low-cost civilian C-UAS.
[6]	Active Radar (Micro-Doppler)	Classification of small UAVs vs. birds using m-D signatures.	Requires high dwell time and fails against hovering drones due to the *Zero-Doppler Notch*.
[11]	Passive RF (Deep Learning)	Identification of drone models via RF fingerprinting using CNNs.	Ineffective against autonomous "Dark Drones" (Radio Silence) and encrypted links.
[12]	Acoustic Sensors (Late Fusion)	Low-cost detection using microphone arrays and audio processing.	Severely limited range ($<$ 150m) and high susceptibility to urban noise pollution.
[13]	Computer Vision (UAV-YOLO)	Optimized YOLO architecture for small object detection.	Suffers from high latency on edge devices without hardware acceleration.
[14]	Multi-Sensor Fusion	Review of fusion architectures combining Radar and Vision.	Focuses on detection only; lacks "Cyber-Physical" integration with UTM for legal interdiction.
[15]	Deep Learning Survey	Comprehensive review of CNN, Transformer and hybrid DL architectures for UAV detection.	Lacks hardware deployment analysis; no embedded edge benchmarks.
[16]	Multi-Modal Sensing Advances	Surveys EO/IR and FMCW radar advances against agile targets.	Limited discussion on swarm scenarios and saturation-based attack vectors.
[17]	BRA-YOLOv10 Vision	Attention-based YOLOv10 for UAV small-target detection; 55.3% mAP@50 on Jetson Orin.	Evaluated on Jetson Orin only; no testing on cost-constrained platforms (e.g., RPi 5).
[18]	Anti-UAV Benchmark Survey	Comprehensive benchmarking of anti-UAV detection, tracking, and interdiction methods.	Focuses on single-target tracking; multi-target swarm scenarios under-represented.
[19]	Multi-Modal DL Fusion for LSS UAVs	DL-based radar-vision fusion for low-slow-small UAV detection with improved accuracy.	Fusion latency analysis not provided; UTM integration not addressed.

Table 2: Benchmark of Object Detection Models on Edge Devices

Model [Ref.]	Input	mAP@50	FPS(FP32)	FPS(INT8)
YOLOv4-T [23]	416	40.2%	32	58
YOLOv7-T [24]	640	51.2%	35	65
YOLOv8n [25]	640	53.9%	50	95
YOLOv10n [17]	640	**55.3%**	**72**	**135**

Note: Performance metrics are synthesized from the respective cited studies. YOLOv10n values represent recent benchmarks on Jetson Orin Nano reported in [17].

Abbreviations:
mAP@50: Mean Average Precision calculated at a 0.5 Intersection over Union (IoU) threshold.
FPS: Frames Per Second (indicates inference speed).
FP32: Floating Point 32-bit (Standard model precision).
INT8: Integer 8-bit (Quantized model precision for edge devices).
Ref.: Reference source citation.

5. Architectural Synthesis

Based on the critical analysis, no single sensor is sufficient. We synthesize a reference architecture that employs a **Discrete Kalman Filter (DKF)** for sensor fusion, an approach validated by recent deep learning-based multi-modal frameworks designed for low-slow-small targets [19].

5.1. The Fusion Pipeline

As illustrated in Fig. 4, the system must integrate Unmanned Traffic Management (UTM) validation. The Radar provides coarse "Cueing" (Range/Azimuth), guiding the PTZ camera to the target area (Slew-to-Cue). The Optical Tracker then provides precise angular data. The Kalman Filter fuses these asynchronous measurements to estimate the state vector, ensuring track continuity even if the drone is temporarily occluded behind obstacles.

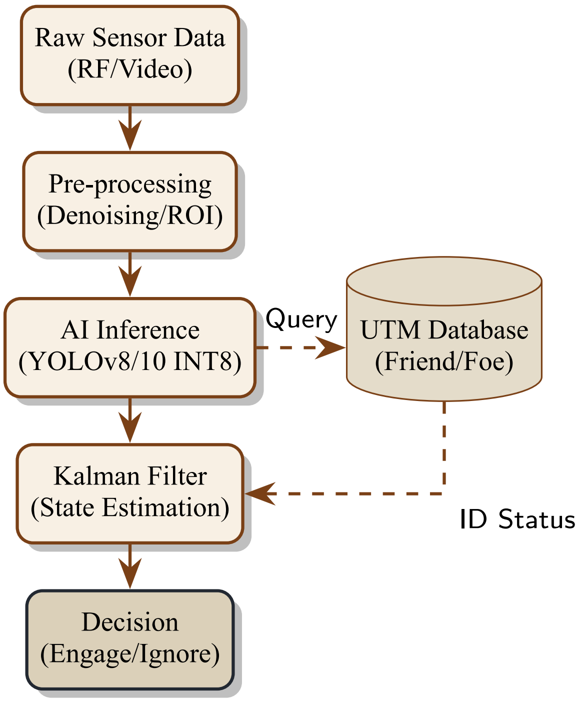

Figure 4: End-to-End Edge Processing Pipeline: The system integrates quantized inference with UTM verification to ensure lawful interdiction.

6. Open Challenges and Future Directions

Despite significant advancements, the C-UAS domain faces several open challenges that future research must address.

6.1. Swarm Attacks and Saturation

Current architectures are largely optimized for Single-Target Tracking (STT). A coordinated swarm attack involving dozens of drones can easily saturate the fusion pipeline and the mechanical slew rate of PTZ cameras. Future research must pivot towards **Multi-Target Tracking (MTT)** algorithms, such as Joint Probabilistic Data Association (JPDA), optimized for parallel execution on edge GPUs as highlighted in recent benchmarks for anti-UAV swarm defense [15, 18].

6.2. The Regulatory Void (UTM Integration)

Neutralizing a drone in a civilian environment carries severe legal risks (e.g., collateral damage from jamming). The integration of **Unmanned Traffic Management (UTM)** databases—as proposed in our architectural synthesis—is not merely a technical feature but a legal necessity [26]. This "Cyber-Physical" verification loop distinguishes compliant delivery drones (Friend) from hostile actors (Foe) before any interdiction occurs.

6.3. Neuromorphic Vision

Traditional frame-based cameras suffer from motion blur at high speeds and low dynamic range. **Event-based cameras (Neuromorphic sensors)**, which report changes in brightness asynchronously with microsecond latency, represent the next frontier for tracking hyper-agile threats, offering a potential solution to the saturation problem of standard vision systems [1].

7. Conclusion

This systematic review confirms that the "Granularity Gap" in modern air defense is a fundamental physical limitation of legacy systems. Bridging this gap requires a paradigm shift towards **Edge-Native Sensor Fusion**. By integrating quantized Deep Learning models with active Radar cueing and UTM validation, modern engineering can create scalable, legally compliant shields against asymmetric threats. The future of C-UAS lies not in larger radars, but in smarter, fused edge intelligence capable of distinguishing a bird from a threat in complex urban environments.

References

[1] A. e. a. Yasmeen, "Recent research progress on ground-to-air vision-based anti-uav detection: A review," *Drones*, 2025.

[2] W. e. a. Khawaja, "A survey on detection, tracking, and classification of aerial threats using radars," *IEEE Comm. Surv. Tut.*, 2022.

[3] V. e. a. Chamola, "A comprehensive review of uav attacks and neutralization techniques," *Ad Hoc Networks*, 2021.

[4] S. e. a. Park, "Survey on anti-drone systems: Components, designs, and challenges," *IEEE Access*, 2021.

[5] M. I. Skolnik, *Introduction to radar systems*. McGraw-hill, 2001.

[6] P. e. a. Molchanov, "Classification of small uavs and birds by micro-doppler signatures," *Int. J. Microw. Wirel. Tech.*, 2014.

[7] I. e. a. Bisio, "Unauthorized amateur uav detection based on wifi fingerprint," *IEEE Comm. Mag.*, 2018.

[8] B. e. a. Wang, "A survey on vision-based anti unmanned aerial vehicles methods," *Drones*, 2024.

[9] M. H. e. a. Rahman, "A comprehensive survey of uav detection and classification using machine learning approaches," *Remote Sensing*, 2024.

[10] X. e. a. Lyu, "A survey on detection and tracking of small drones," *IEEE Access*, 2023.

[11] M. F. e. a. Al-Sa'd, "Rf-based drone detection and identification using deep learning," *Future Gen. Comp. Sys.*, 2019.

[12] P. e. a. Casabianca, "Acoustic-based uav detection using late fusion," *Drones*, 2021.

[13] M. e. a. Liu, "Uav-yolo: Small object detection on uav perspective," *Sensors*, 2020.

[14] J. e. a. Li, "Multi-sensor fusion for uav detection: A review," *IEEE Access*, 2022.

[15] N. e. a. Al-Qubaydhi, "Deep learning for uavs detection: A review," *Comp. Sci. Rev.*, 2024.

[16] U. e. a. Seidaliyeva, "Advances and challenges in drone detection," *Sensors*, 2024.

[17] Q. Zhang, X. Wang *et al.*, "Bra-yolov10: Uav small target detection based on yolov10," *Drones*, no. 3, p. 159, 2025.

[18] J. Dong *et al.*, "Securing the skies: A comprehensive survey on anti-uav methods, benchmarking, and future directions," in *CVPR Workshops*, 2025.

[19] Y. Li *et al.*, "Research on multi-modal fusion detection method for low-slow-small uavs based on deep learning," *Drones*, no. 12, 2024.

[20] NVIDIA, "Deep learning inference optimization with tensorrt," *Technical Whitepaper*, 2023.

[21] X. e. a. Zhang, "Lightweight neural networks for edge devices," *IEEE Trans. Consum. Electron.*, 2022.

[22] Y. e. a. Chen, "Edge computing for real-time uav detection," *IEEE IoT J.*, 2023.

[23] A. e. a. Bochkovskiy, "Yolov4: Optimal speed and accuracy of object detection," in *arXiv*, 2020.

[24] C.-Y. e. a. Wang, "Yolov7: Trainable bag-of-freebies sets new state-of-the-art," in *CVPR*, 2023.

[25] G. e. a. Jocher, "Yolo by ultralytics," *GitHub*, 2023.

[26] O. e. a. Aouedi, "Intelligent traffic management for uavs," *IEEE Trans. Intell. Transp. Sys.*, 2021.

Specific Roles of Authors

Islam Emad Ismail: Conceptualization, Methodology, Writing - Original Draft, Data Curation, Formal Analysis, Visualization.
Rawan, Sagy, Abdelaziz, Rafeek, Islam Hesham: Validation, Investigation, Data Collection, Reviewing.
Rania Hamdy Elabd, Mervat El-Seddek, Amr Hussein Abdallah: Supervision, Writing - Review & Editing, Project Administration.

Conflict of Interest

The authors report no financial or personal interests that could have biased the research presented in this paper.

Disclosure of Generative AI and AI Tools Used in Writing

The authors declare the use of generative AI tools strictly for linguistic proofreading and formatting the document layout to comply with the journal's template requirements. No AI tools were used to fabricate the core analytical findings presented in this study.